

TIANHAO CAO

British Columbia | Email | Website | LinkedIn

OBJECTIVE

M.S. Data Science candidate (UBC, Computational Linguistics) with broad expertise spanning LLM fine-tuning, RAG pipelines, end-to-end machine learning, causal inference, and production AI deployment. Passionate about building systems where research rigor meets engineering discipline — from privacy-aware LLM research to scalable, cloud-hosted AI services.

EDUCATION

University of British Columbia

Sep 2025 – Jun 2026

M.S. in Data Science (Computational Linguistics)

Vancouver, BC

Relevant Coursework: Supervised Machine Learning, Unsupervised Machine Learning, Deep Learning, Natural Language Processing, Feature & Model Selection, Data Visualization, Statistical Inference & Computation, Databases & Data Retrieval, Time Series Analysis, Machine Translation, Computational Semantics, Advanced Corpus Linguistics

University of California, Santa Cruz

Sep 2019 – Jun 2022

B.A. in Business Management Economics

Santa Cruz, CA

Relevant Coursework: Machine Learning, Econometrics, Advanced Quantitative Methods, Statistics, Corporate Finance, Marketing

TECHNICAL SKILLS

ML & AI

PyTorch, Scikit-learn, XGBoost, Hugging Face Transformers, LLM Fine-tuning, RAG Pipelines, Generative AI, Vector Embeddings, Prompt Engineering, SHAP (Model Explainability)

MLOps & Cloud

Docker, AWS, Git / GitHub, CI/CD, Weights & Biases (W&B)

Programming

Python (primary), R, SQL, MongoDB, Bash

Data Engineering

ETL Pipelines, Feature Engineering, Pandas, NumPy, FastAPI, Data Visualization

Languages

English (Native), Mandarin (Native)

PROJECTS

Membership Inference Attack on Fine-tuned Language Models

Academic Project – Kaggle Competition

GitHub Repository

- Achieved **AUC of 0.907** and TPR@FPR=0.1 of 0.697 — a **+55% AUC improvement** and **+349% TPR improvement** over the raw-loss baseline — by designing an Improved Neighborhood Attack using dual-model feature fusion across four LLM variants (SmolLM2 135M and 360M) with PyTorch and Hugging Face Transformers.
- Engineered a 16-dimensional feature vector spanning fine-tuned model losses, base model reference ratios, neighborhood perturbation gaps, and cross-model agreement signals, enabling a Gradient Boosting classifier (300 estimators) to learn nuanced LLM memorization patterns.
- Replaced slow BERT mask-and-fill neighbor generation with a fast token-drop perturbation strategy (10% word dropout, 5 neighbors per sample), dramatically reducing inference time while maintaining perturbation quality.
- Revealed through feature importance analysis that cross-model reference signals account for 86% of memorization detection importance, surfacing the key insight that larger fine-tuned models produce stronger and more detectable memorization signatures.

Full-Stack Corpus Search Engine with LLM Annotation

University of British Columbia

Academic Project

In Progress

- Reduced manual labeling effort by 80%+ by fine-tuning a domain-specific NLP model on Hugging Face Transformers for automated corpus annotation, maintaining production-grade annotation quality at scale.

- Designed and developed a FastAPI backend with Whoosh-based retrieval indexing, enabling sub-second keyword queries across large text corpora, and deployed the containerized application to AWS using Docker for reproducible, scalable cloud access.
- Automated dataset acquisition via the Kaggle API and built structured ETL workflows to prepare raw text data for indexing and annotation, implementing RESTful endpoints with structured error handling and input validation.

Credit Card Default Analysis

University of British Columbia

Academic Project

GitHub Repository

- Achieved the best precision-recall balance (F1: 0.55) on a heavily imbalanced 30,000-sample dataset by building an end-to-end ML pipeline — conducting EDA, applying resampling and feature scaling, and training Logistic Regression, Random Forest, and XGBoost classifiers.
- Applied SHAP (SHapley Additive exPlanations) analysis to interpret the optimized XGBoost model, identifying recent payment history as the most critical predictor of default and translating black-box decisions into actionable, non-technical insights.

Predictive Modeling of Agricultural Insurance Purchase Behavior

UC Santa Cruz

Academic Project

GitHub Repository

- Achieved 76% predictive precision in modeling farmers' insurance purchase behavior by developing and comparing Lasso Regression and Random Forest models on a 1,000+ record behavioral survey dataset (30+ variables) from the Harvard Dataverse.
- Interpreted feature importance outputs to identify key behavioral drivers of insurance adoption, translating technical findings into actionable business recommendations and conducting error analysis to assess model limitations and potential data biases.

Difference-in-Differences Analysis of COVID-19 Policy Impacts

UC Santa Cruz

Academic Project

GitHub Repository

- Quantified the causal impact of early pandemic lockdown policies on GDP across 5 major economies (US, China, Japan, UK, India) by implementing a Difference-in-Differences (DiD) framework on cross-country economic panel data from the WHO and Our World in Data.
- Mitigated omitted-variable bias by incorporating control variables and performed robustness checks against baseline models, validating causal estimates and surfacing potential confounding factors in policy analysis.